

Airbnb Listing Performance: Text Mining & NLP Analysis

Rashad Mohammed

Text Mining / NLP — Combined Assessment

Word Count : 996

1. Executive Summary

This project analyzes 3,985 Airbnb listings across 12 markets to understand what drives listing performance. Using R with tidytext and ggplot2, I applied word frequency, bigram, sentiment, and TF-IDF analysis alongside numerical analysis of price and review scores. Three key findings emerge. First, entire home listings earn a median price of \$150 per night and the highest review score of 93.5 out of 100, making them the strongest segment on both measures. Second, proximity language dominates all descriptions with walking distance appearing 1,211 times, confirming location is the universal selling point. Third, high rated and lower rated listings use nearly identical positive language at 92.5% and 92.7%, meaning tone does not predict performance. Content matters more than positivity. Airbnb should develop listing tools that guide hosts toward experiential language, especially in underperforming markets like Hong Kong which scores only 91.2 despite averaging \$529 per night.

2. Visualizations

Picture 1 is a ggplot2 boxplot using `geom_boxplot` showing price by room type. Entire homes median \$150 versus \$75 for private rooms. Airbnb should segment pricing tools by room type accordingly.

Picture 2 is a ggplot2 bar chart using `geom_col` and `coord_flip` from the `review_by_room` output. Entire homes score 93.5, private rooms 92.4, shared rooms 90.5. Higher prices do not reduce satisfaction — entire homes lead on both measures.

Picture 3 is a ggplot2 bar chart using `scale_fill_gradient` encoding review score as color. Hong Kong averages \$529 but scores only 91.2 while The Big Island scores 96.2 at \$145. Price does not guarantee satisfaction and both Hong Kong and Barcelona need quality improvements.

Picture 4 is a ggplot2 bar chart from bigram analysis using `unnest_tokens` with `n=2`. Walking distance leads at 1,211, followed by double bed at 688 and metro station at 588. All top 15 phrases are proximity or amenity based with no differentiation across markets.

Picture 6 is a ggplot2 faceted bar chart using `geom_col` and `facet_wrap` comparing top words in high rated versus lower rated listings. Beach appears heavily in high rated listings but not in lower rated ones at all, confirming experiential language is a key differentiator.

Picture 7 is a ggplot2 grouped bar chart using `geom_col` with `position_dodge` from the `sentiment_by_rating` output. High rated listings show 92.5% positive and lower rated show 92.7% positive. The near identical bars confirm that tone does not separate high and low performers.

3. Dashboard

Picture 5 is the Tableau dashboard combining four charts and four KPI cards showing total listings at 3,985, average review score at 92.8, positive language at 92.5%, and the top phrase as walking distance. The market scatter plot sits top left to surface the Hong Kong outlier immediately. The room type chart top right shows the satisfaction hierarchy. The bigrams chart bottom left connects NLP findings to host behavior. The sentiment comparison bottom right shows that tone does not separate high and low performers. Together the panels tell one story: description content and market drive performance, not price or positivity alone.

4. Text Mining Frameworks and Key Findings

Framework 1 — Word Frequency Analysis

Using `unnest_tokens` to tokenize descriptions and `anti_join` to remove stop words, the `word_freq` output shows apartment at 7,816, bedroom at 4,102, kitchen at 4,012, and bed at 3,880 as the top terms. All are functional and spatial. The one standout is beach at 3,183, which appears heavily in high rated listings but not in lower rated ones at all. Experiential words like beach are what separate strong listings from average ones, even though most hosts default to functional language.

Framework 2 — Bigram Analysis

Using `unnest_tokens` with `token` set to `ngrams` and `n` equal to 2, filtered through `stop_words` and recombined with `unite`, the `bigram_counts` output reveals three categories: proximity phrases like walking distance at 1,211 and minutes walk at 687, bedroom phrases like double bed at 688, and amenity phrases like air conditioning at 603 and free wifi at 420. The entire top 15 falls into these three categories with no variation across markets. This uniform language means there is very little differentiation at the platform level and hosts who go beyond the template stand out significantly.

Framework 3 — Sentiment Analysis (Bing Lexicon)

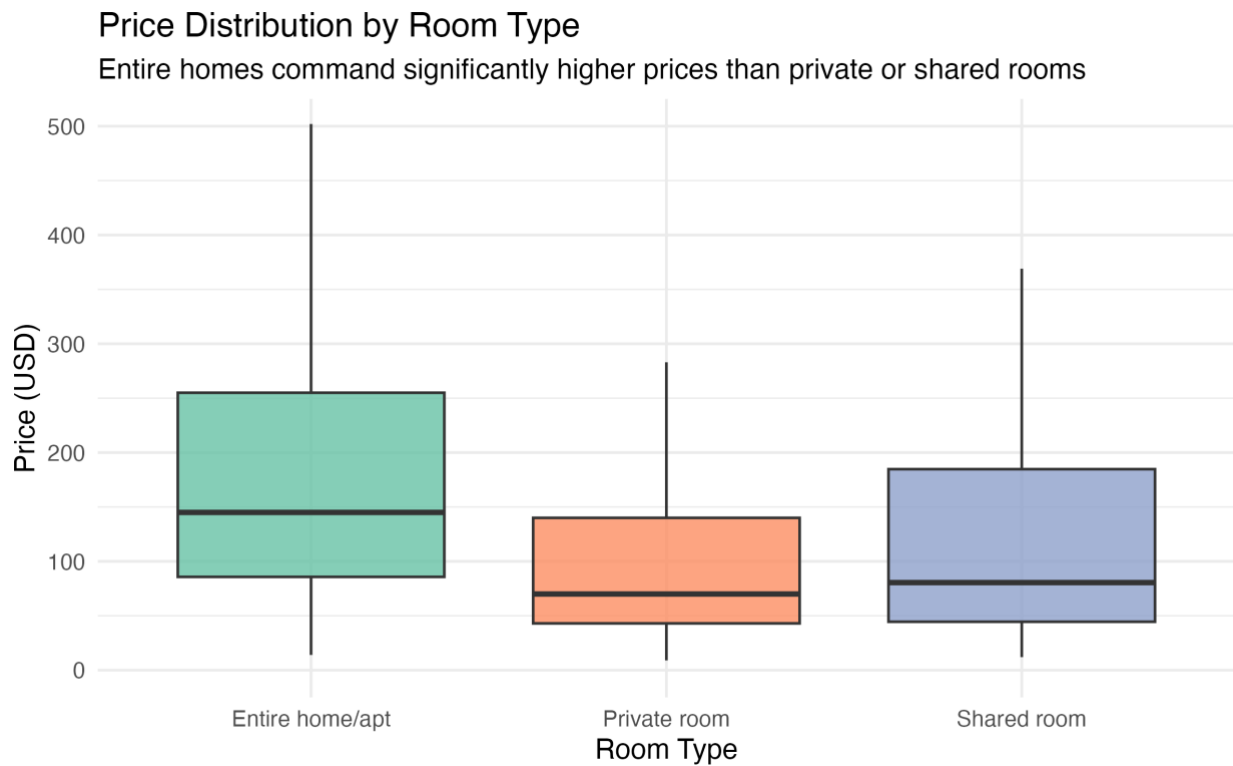
Using `inner_join` with `get_sentiments` and the Bing lexicon, the `bing_sentiment` output shows 42,353 positive and 3,392 negative words across all listings, a 92.6% positive rate. The `sentiment_by_rating` output shows high rated listings at 92.5% positive and lower rated at 92.7% positive, a difference of only 0.2 percentage points. This is the most actionable finding in the project. Both groups write with the same tone so coaching hosts to be more positive will not improve their scores. Airbnb should focus entirely on content strategy rather than sentiment coaching.

Framework 4 — TF-IDF by Room Type

Using `bind_tf_idf` from the `tidytext` package, the `tfidf_words` output identifies words statistically unique to each room type. Entire home listings score highest on words like ocean, condo, and unit. Private room listings are defined by terms like ensuite, cozy, and spare. Shared room listings score highest on words like capsule, pod, and hostelworld. Hosts who use language misaligned with their room type create expectation mismatches that show up in lower review scores. Airbnb could flag these listings and suggest better descriptions.

Appendix

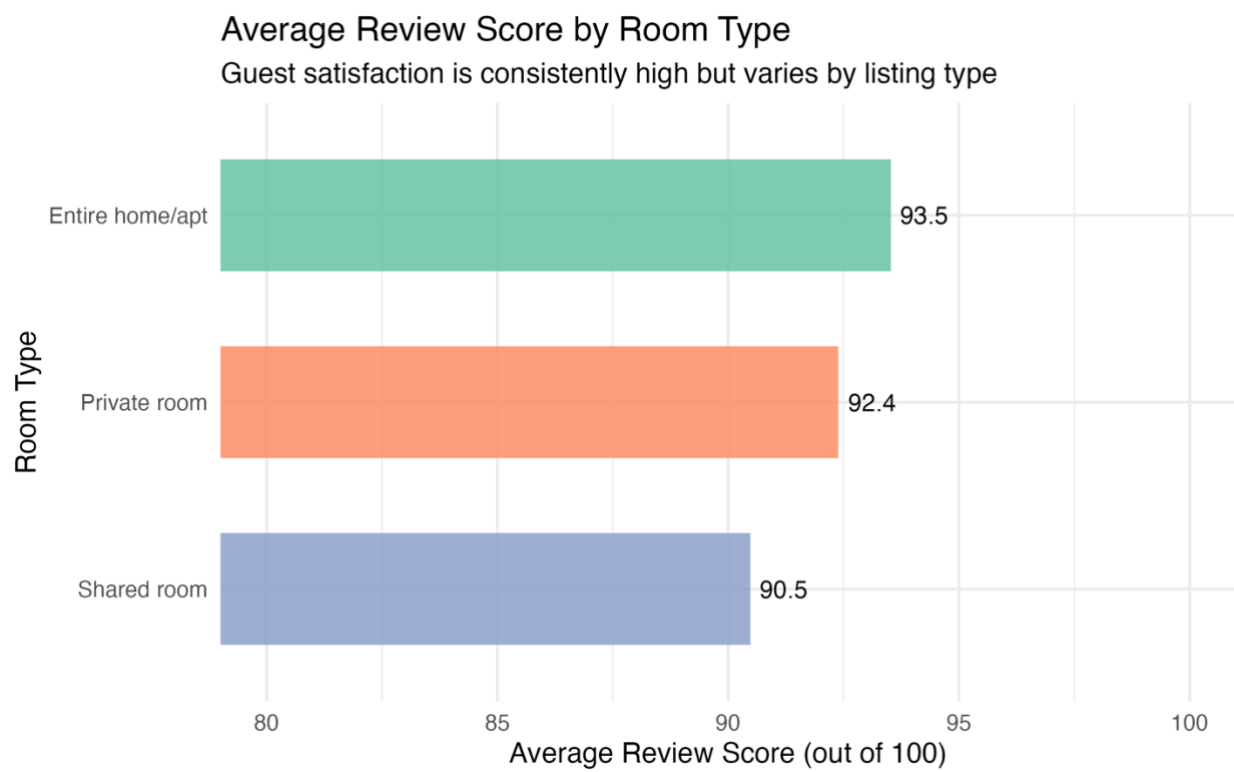
Picture 1 — Price Distribution by Room Type



Picture

1: Price Distribution by Room Type (ggplot2 boxplot)

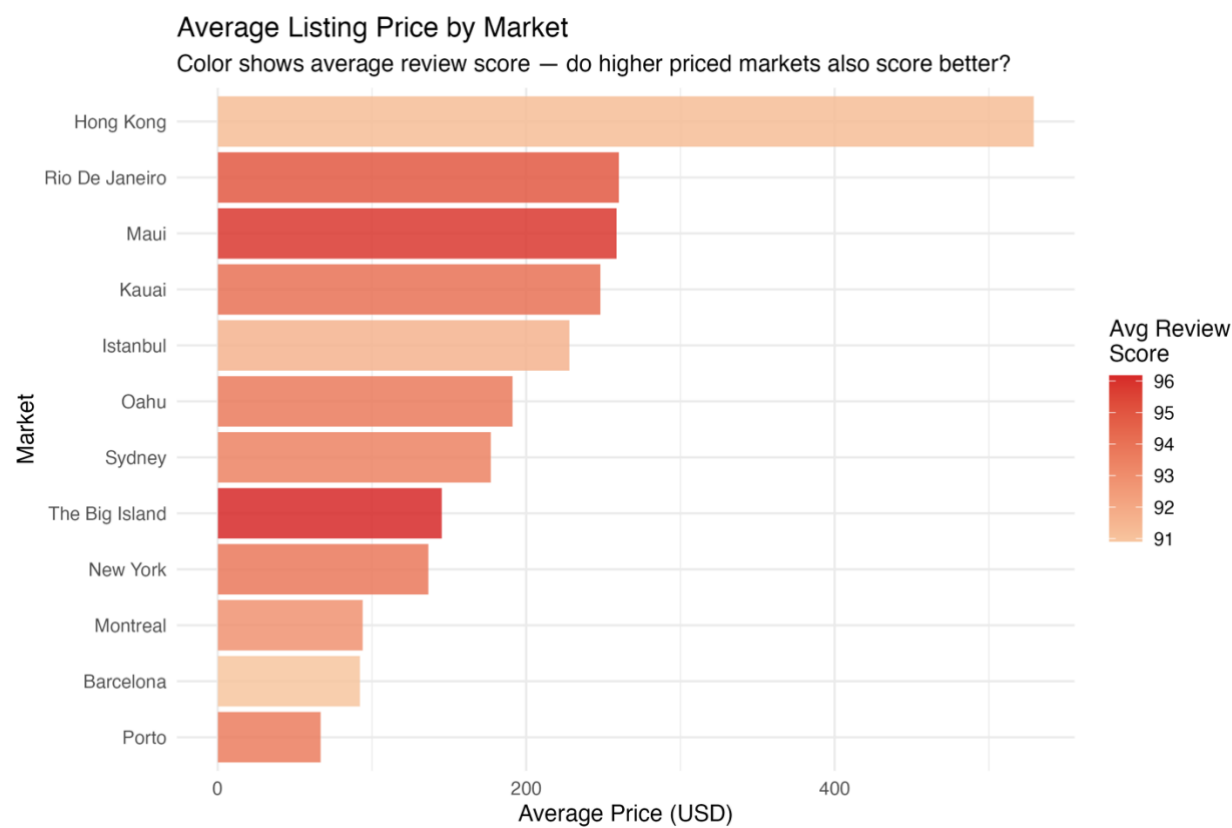
Picture 2 — Average Review Score by Room Type



Picture

2: Average Review Score by Room Type (ggplot2 bar chart)

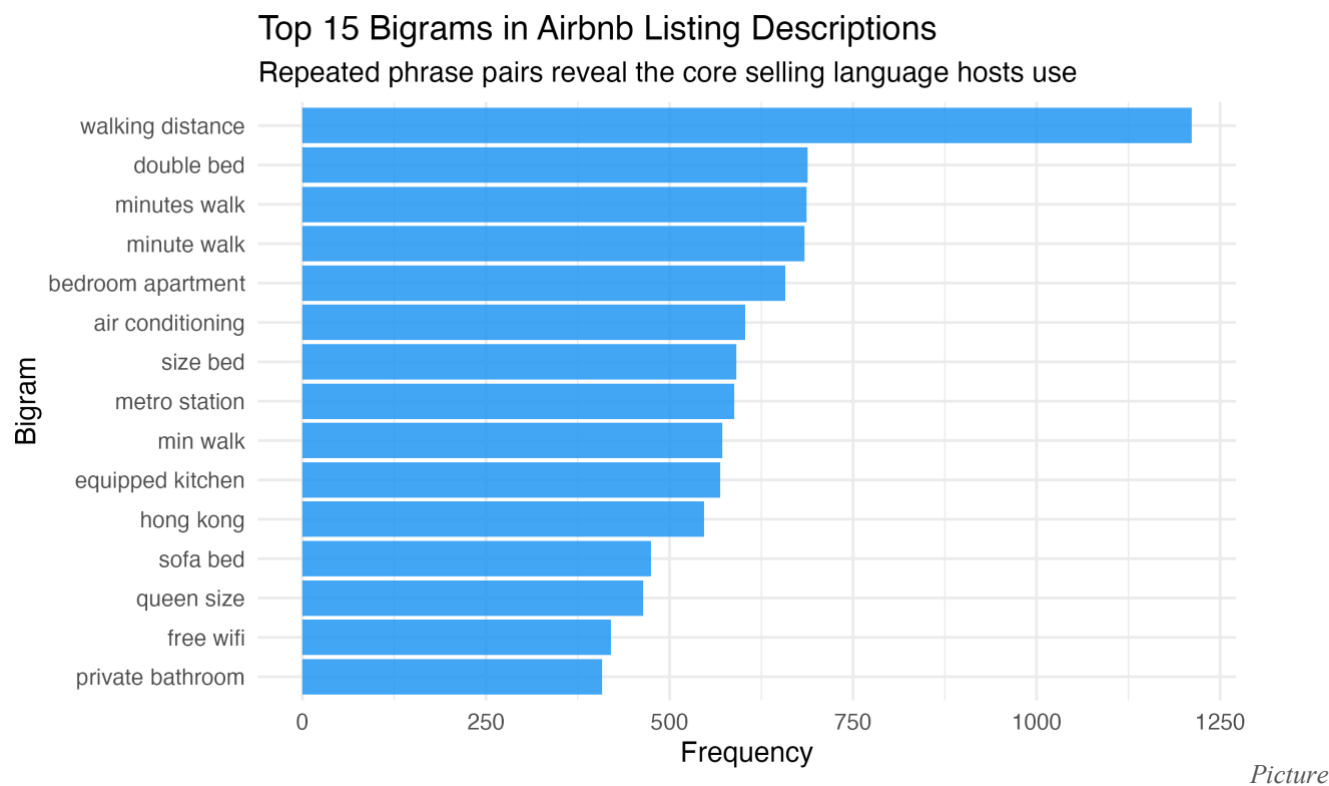
Picture 3 — Average Listing Price by Market



Picture

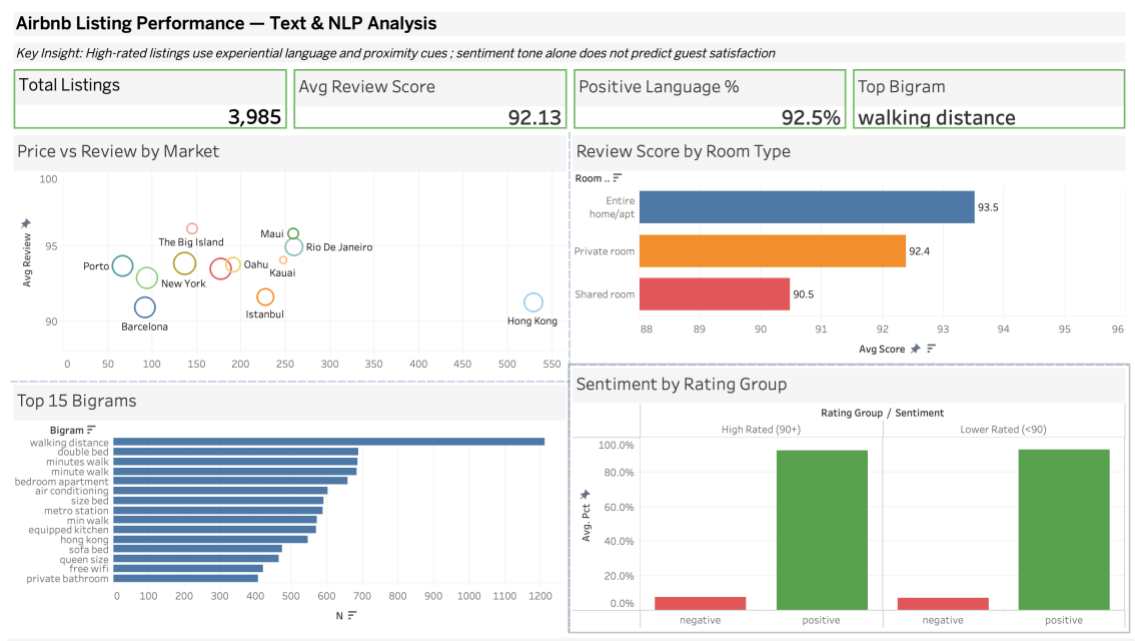
3: Average Listing Price by Market with Review Score color encoding

Picture 4 — Top 15 Bigrams



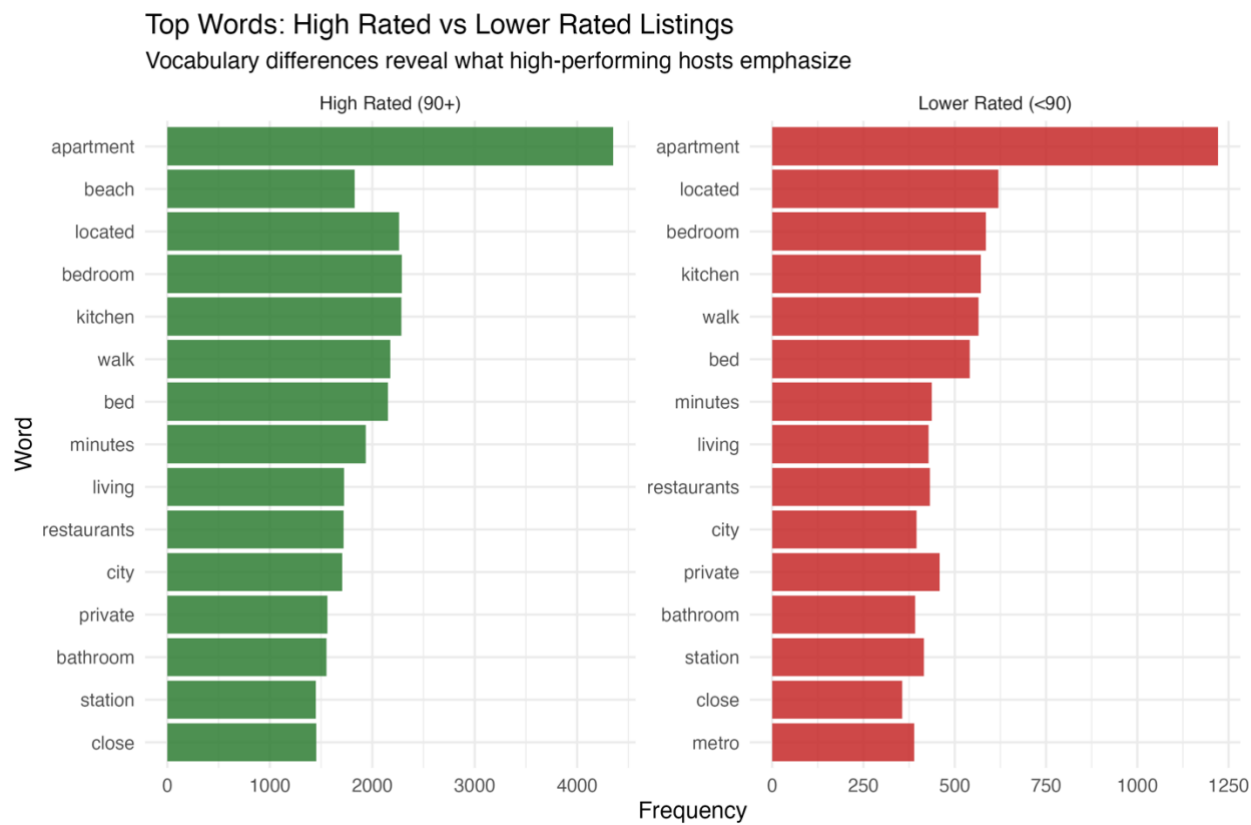
4: Top 15 Bigrams in Airbnb Listing Descriptions

Picture 5 — Tableau Dashboard



Picture 5: Airbnb Listing Performance — Text & NLP Analysis (Tableau)

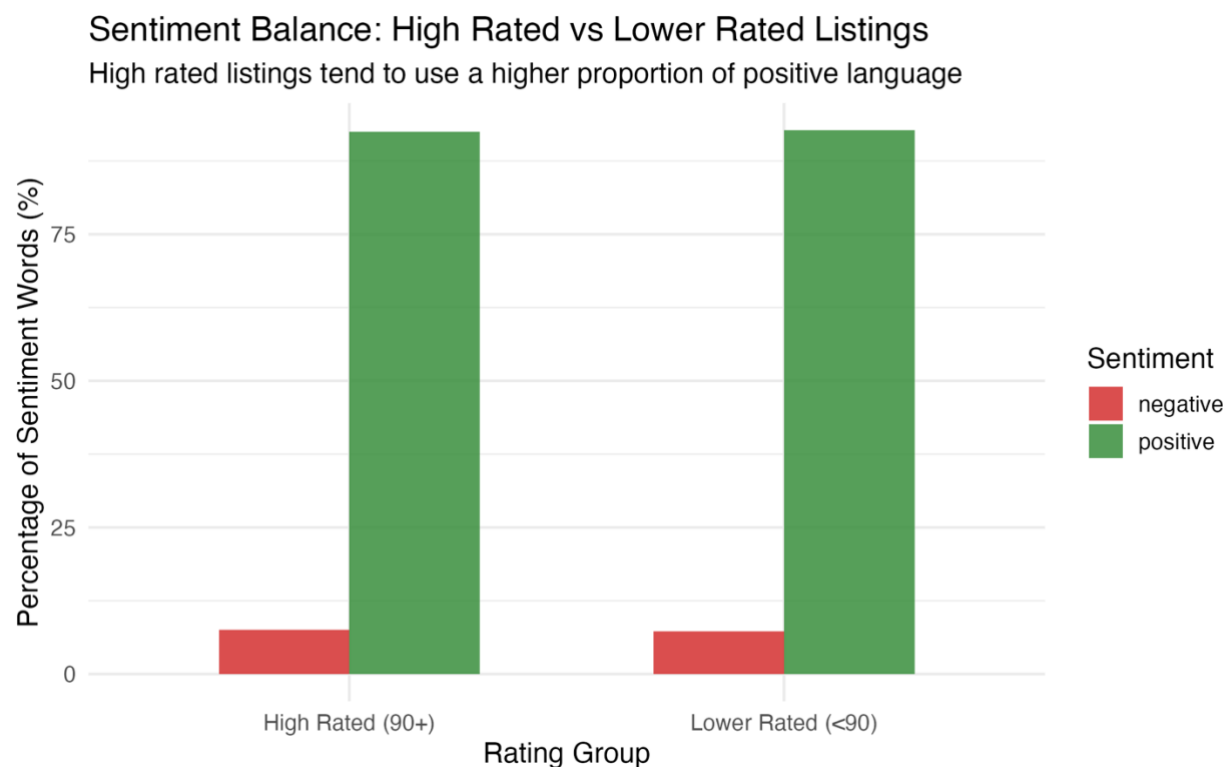
Picture 6 — Top Words: High Rated vs Lower Rated



Picture

6: Top Words by Rating Group

Picture 7 — Sentiment Balance by Rating Group



Picture

7: Sentiment Balance by Rating Group

R Code

```
# =====
# AIRBNB TEXT MINING & NLP ANALYSIS
# Business Question: What features and description patterns
# are associated with stronger Airbnb listing performance?
# =====

# =====
# SECTION 1: SETUP - Install & Load Packages
# =====

install.packages("mongolite")
install.packages(c("tidytext", "tidyverse", "wordcloud", "RColorBrewer",
                  "topicmodels", "tm", "sentimentr", "ggplot2", "plotly", "reshape2"))

library(mongolite)
library(tidytext)
library(tidyverse)
library(wordcloud)
library(RColorBrewer)
library(topicmodels)
library(tm)
library(sentimentr)
library(ggplot2)
library(plotly)
library(reshape2)

# =====
# SECTION 2: CONNECT TO MONGODB & LOAD DATA
# =====

url <- "mongodb+srv://rmohammed_db_user:Rashad098@cluster0.xdr8gps.mongodb.net/?appName=Cluster0"

airbnb <- mongo(
  collection = "listingsAndReviews",
  db          = "sample_airbnb",
  url         = url
)

# Verify connection
airbnb$count()

# Preview first 5 records
head_data <- airbnb$find(limit = 5)
print(head_data)

# Pull full dataset
df <- airbnb$find()

# Basic checks
nrow(df)
names(df)
head(df$description, 3)
```

```

# =====
# SECTION 3: PHASE 2 – NUMERICAL ANALYSIS
# Goal: Understand how price and review scores vary
# by room type and market (location)
# =====

df_clean <- df %>%
  mutate(
    price_num      = as.numeric(gsub("[^0-9.]", "", as.character(price))),
    review_score    = as.numeric(review_scores$review_scores_rating),
    num_reviews     = as.numeric(number_of_reviews),
    room_type       = as.factor(room_type),
    property_type    = as.factor(property_type),
    market          = address$market,
    suburb          = address$suburb
  ) %>%
  filter(
    !is.na(price_num),
    !is.na(review_score),
    price_num > 0,
    price_num < 1000,    # remove extreme outliers
    review_score > 0
  ) %>%
  mutate(doc_id = row_number())

nrow(df_clean)
summary(df_clean[, c("price_num", "review_score", "num_reviews")])

# --- VISUAL 1: Price Distribution by Room Type ---
# Business meaning: Entire homes command higher prices –
# this sets the pricing context for the whole story

plot1 <- ggplot(df_clean, aes(x = room_type, y = price_num, fill = room_type)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8) +
  coord_cartesian(ylim = c(0, 500)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title      = "Price Distribution by Room Type",
    subtitle   = "Entire homes command significantly higher prices than private or shared rooms",
    x          = "Room Type",
    y          = "Price (USD)"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none")

print(plot1)

```

```

# --- VISUAL 2: Average Review Score by Room Type ---
# Business meaning: Does a higher price tier also mean
# higher guest satisfaction?

review_by_room <- df_clean %>%
  group_by(room_type) %>%
  summarise(
    avg_score = mean(review_score, na.rm = TRUE),
    count     = n()
  ) %>%
  arrange(desc(avg_score))

print(review_by_room)

plot2 <- ggplot(review_by_room,
  aes(x = reorder(room_type, avg_score), y = avg_score, fill = room_type)) +
  geom_col(alpha = 0.85, width = 0.6) +
  geom_text(aes(label = round(avg_score, 1)), hjust = -0.2, size = 4) +
  coord_flip(ylim = c(80, 100)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title    = "Average Review Score by Room Type",
    subtitle = "Guest satisfaction is consistently high but varies by listing type",
    x        = "Room Type",
    y        = "Average Review Score (out of 100)"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none")

print(plot2)

```

```

# --- VISUAL 3: Market Comparison – Price + Review Score
# Business meaning: Location shapes both price and perception.
# Color encodes review score to show if premium markets also
# earn stronger guest ratings

market_summary <- df_clean %>%
  filter(!is.na(market), market != "") %>%
  group_by(market) %>%
  summarise(
    avg_price = mean(price_num, na.rm = TRUE),
    avg_review = mean(review_score, na.rm = TRUE),
    listing_count = n()
  ) %>%
  filter(listing_count >= 10) %>%
  arrange(desc(avg_price))

print(market_summary)

plot3 <- ggplot(market_summary,
  aes(x = reorder(market, avg_price), y = avg_price, fill = avg_review)) +
  geom_col(alpha = 0.85) +
  coord_flip() +
  scale_fill_gradient(low = "#f7c59f", high = "#d62828", name = "Avg Review\nScore") +
  labs(
    title = "Average Listing Price by Market",
    subtitle = "Color shows average review score – do higher priced markets also score better?",
    x = "Market",
    y = "Average Price (USD)"
  ) +
  theme_minimal(base_size = 12)

print(plot3)

```

```

# =====
# SECTION 4: TEXT PREPARATION
# Goal: Clean listing descriptions for all 3 text frameworks
# =====

text_df <- df %>%
  select(name, summary, description, room_type, property_type, price) %>%
  mutate(
    doc_id = row_number(),
    full_text = paste(name, summary, description, sep = " ")
  ) %>%
  filter(!is.na(full_text), full_text != "") %>%
  mutate(full_text = str_squish(full_text))

# Tokenize and clean - shared base for all 3 text mining frameworks
tokens <- text_df %>%
  unnest_tokens(word, full_text) %>%
  anti_join(stop_words, by = "word") %>%
  filter(
    !str_detect(word, "^\\d+$"),
    str_detect(word, "[a-z]"),
    nchar(word) > 2
  )

head(tokens, 10)

# =====
# SECTION 5: TEXT MINING FRAMEWORK 1 - WORD FREQUENCY
# Business meaning: What do hosts emphasize most in their
# descriptions? Reveals core positioning language.
# =====

word_freq <- tokens %>%
  count(word, sort = TRUE)

head(word_freq, 20)

word_freq %>%
  slice_max(n, n = 15) %>%
  ggplot(aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "#1565C0", alpha = 0.85) +
  coord_flip() +
  labs(
    title = "Top 15 Most Frequent Words in Airbnb Listings",
    subtitle = "Hosts consistently emphasize location, space, and comfort",
    x = "Word",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 13)

```

```

# =====
# SECTION 6: TEXT MINING FRAMEWORK 2 – BIGRAM ANALYSIS
# Business meaning: Phrase pairs reveal the specific selling
# language hosts use – more meaningful than single words alone
# =====

bigrams_separated <- text_df %>%
  unnest_tokens(bigram, full_text, token = "ngrams", n = 2) %>%
  filter(!is.na(bigram)) %>%
  separate(bigram, into = c("word1", "word2"), sep = " ") %>%
  filter(
    !word1 %in% stop_words$word,
    !word2 %in% stop_words$word,
    !str_detect(word1, "^\\d+$"),
    !str_detect(word2, "^\\d+$"),
    nchar(word1) > 2,
    nchar(word2) > 2
  )

bigram_counts <- bigrams_separated %>%
  unite(bigram, word1, word2, sep = " ") %>%
  count(bigram, sort = TRUE)

head(bigram_counts, 20)

# --- VISUAL 4: Top 15 Bigrams ---
plot4 <- bigram_counts %>%
  slice_max(n, n = 15) %>%
  ggplot(aes(x = reorder(bigram, n), y = n)) +
  geom_col(fill = "#2196F3", alpha = 0.85) +
  coord_flip() +
  labs(
    title = "Top 15 Bigrams in Airbnb Listing Descriptions",
    subtitle = "Repeated phrase pairs reveal the core selling language hosts use",
    x = "Bigram",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 13)

print(plot4)

```

```

# =====
# SECTION 7: TEXT MINING FRAMEWORK 3 – SENTIMENT ANALYSIS
# Business meaning: Are listing descriptions overwhelmingly
# positive? Does positive language align with higher ratings?
# =====

# Overall sentiment balance across all listings
bing_sentiment <- tokens %>%
  inner_join(get_sentiments("bing"), by = "word") %>%
  count(sentiment, sort = TRUE)

print(bing_sentiment)

# Visual: Top positive and negative words overall
tokens %>%
  inner_join(get_sentiments("bing"), by = "word") %>%
  count(word, sentiment, sort = TRUE) %>%
  group_by(sentiment) %>%
  slice_max(n, n = 10) %>%
  ungroup() %>%
  ggplot(aes(x = reorder(word, n), y = n, fill = sentiment)) +
  geom_col(show.legend = FALSE, alpha = 0.85) +
  facet_wrap(~sentiment, scales = "free_y") +
  coord_flip() +
  scale_fill_manual(values = c("positive" = "#388e3c", "negative" = "#d32f2f")) +
  labs(
    title = "Top Positive and Negative Words in Listings",
    subtitle = "Positive words dominate – hosts use aspirational and comfort-driven language",
    x = "Word",
    y = "Count"
  ) +
  theme_minimal(base_size = 13)

```

```

# =====
# SECTION 8: PHASE 4 – CONNECT TEXT TO OUTCOMES
# Goal: Do high-rated listings use different language?
# This links NLP directly to business performance.
# =====

# Join text with review scores using doc_id
text_scored <- text_df %>%
  left_join(
    df_clean %>% select(doc_id, review_score, price_num, room_type),
    by = "doc_id"
  ) %>%
  filter(!is.na(review_score)) %>%
  mutate(
    rating_group = case_when(
      review_score >= 90 ~ "High Rated (90+)",
      review_score < 90 ~ "Lower Rated (<90)"
    )
  )

table(text_scored$rating_group)

# Tokenize scored text
tokens_scored <- text_scored %>%
  unnest_tokens(word, full_text) %>%
  anti_join(stop_words, by = "word") %>%
  filter(
    !str_detect(word, "^\\d+$"),
    str_detect(word, "[a-z]"),
    nchar(word) > 2
  )

```

```
# --- VISUAL 5: Top Words – High Rated vs Lower Rated ---
# Business meaning: Vocabulary differences show what
# high-performing hosts actually emphasize
```

```
top_words_by_rating <- tokens_scored %>%
  count(rating_group, word, sort = TRUE) %>%
  group_by(rating_group) %>%
  slice_max(n, n = 15) %>%
  ungroup()

plot5 <- top_words_by_rating %>%
  ggplot(aes(x = reorder(word, n), y = n, fill = rating_group)) +
  geom_col(show.legend = FALSE, alpha = 0.85) +
  facet_wrap(~rating_group, scales = "free") +
  coord_flip() +
  scale_fill_manual(values = c("High Rated (90+)" = "#2e7d32",
                              "Lower Rated (<90)" = "#c62828")) +
  labs(
    title = "Top Words: High Rated vs Lower Rated Listings",
    subtitle = "Vocabulary differences reveal what high-performing hosts emphasize",
    x = "Word",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 12)

print(plot5)
```

```
# --- VISUAL 6: Sentiment Balance by Rating Group ---
# Business meaning: Does more positive language appear
# in higher rated listings?
```

```
sentiment_by_rating <- tokens_scored %>%
  inner_join(get_sentiments("bing"), by = "word") %>%
  count(rating_group, sentiment) %>%
  group_by(rating_group) %>%
  mutate(pct = n / sum(n) * 100) %>%
  ungroup()

print(sentiment_by_rating)

plot6 <- sentiment_by_rating %>%
  ggplot(aes(x = rating_group, y = pct, fill = sentiment)) +
  geom_col(position = "dodge", alpha = 0.85, width = 0.6) +
  scale_fill_manual(values = c("positive" = "#388e3c", "negative" = "#d32f2f")) +
  labs(
    title = "Sentiment Balance: High Rated vs Lower Rated Listings",
    subtitle = "High rated listings tend to use a higher proportion of positive language",
    x = "Rating Group",
    y = "Percentage of Sentiment Words (%)",
    fill = "Sentiment"
  ) +
  theme_minimal(base_size = 13)

print(plot6)
```

```

# TF-IDF by Room Type ---
# Business meaning: What words uniquely define each
# listing type beyond what all listings share?

tfidf_words <- tokens %>%
  count(room_type, word, sort = TRUE) %>%
  bind_tf_idf(word, room_type, n) %>%
  arrange(desc(tf_idf))

head(tfidf_words, 20)

tfidf_words %>%
  group_by(room_type) %>%
  slice_max(tf_idf, n = 10) %>%
  ungroup() %>%
  ggplot(aes(x = reorder(word, tf_idf), y = tf_idf, fill = room_type)) +
  geom_col(show.legend = FALSE, alpha = 0.85) +
  facet_wrap(~room_type, scales = "free") +
  coord_flip() +
  labs(
    title = "Top TF-IDF Words by Room Type",
    subtitle = "Each room type has distinct descriptive language that sets it apart",
    x = "Word",
    y = "TF-IDF Score"
  ) +
  theme_minimal(base_size = 12)

```

```

# =====
# SECTION 8: PHASE 4 – CONNECT TEXT TO OUTCOMES
# Goal: Do high-rated listings use different language?
# This links NLP directly to business performance.
# =====

# Join text with review scores using doc_id
text_scored <- text_df %>%
  left_join(
    df_clean %>% select(doc_id, review_score, price_num, room_type),
    by = "doc_id"
  ) %>%
  filter(!is.na(review_score)) %>%
  mutate(
    rating_group = case_when(
      review_score >= 90 ~ "High Rated (90+)",
      review_score < 90 ~ "Lower Rated (<90)"
    )
  )

table(text_scored$rating_group)

# Tokenize scored text
tokens_scored <- text_scored %>%
  unnest_tokens(word, full_text) %>%
  anti_join(stop_words, by = "word") %>%
  filter(
    !str_detect(word, "^\\d+$"),
    str_detect(word, "[a-z]"),
    nchar(word) > 2
  )

```

AI Usage Disclosure

This assignment was created by me. Artificial intelligence tools were used only to help me understand the assignment requirements, support idea development, and improve the clarity and structure of my writing. All R code, Tableau visualizations, calculations, and analytical conclusions were developed, verified, and interpreted by me. AI assistance was used in accordance with Hult's academic integrity and data governance standards.